

Assessing Flood Vulnerability in New York City: Case Studies and 311 Service Data

Who is most affected by flooding in New York City? How are flood risk and flood vulnerability distributed?

By Fiona Dubay



Iconic Crown Victoria yellow taxi cabs in Hoboken, NJ, submerged in Superstorm Sandy's floodwaters.

Photo: <https://www.nj.com>

BACKGROUND

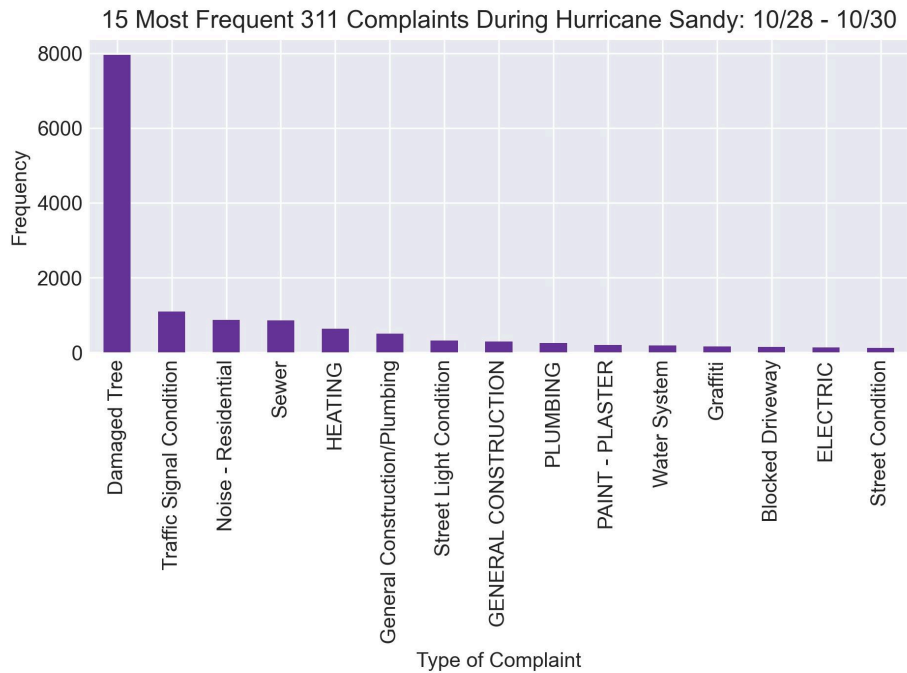
It's well documented that climate change is exacerbating flooding all over the world. As sea levels are rising and more intense rainfall and storm surges are occurring, more people are being put at risk. Flood risk, like all other climate impacts, is not equitably distributed among the world's population. New York City is no exception. During the wet months of the year, a growing portion of New York City's population is finding themselves underwater. Extreme weather events, such as a cloudburst that impacted New York this past July, can bring large amounts of precipitation that have the ability to inundate houses, streets, and entire communities. These extreme weather case study events can be used in order to better understand, and eventually remedy, the issue of distributive flood risk inequity. Using a combination of these weather

events and 311 service request data yields interesting insights into who is experiencing flooding in New York City.

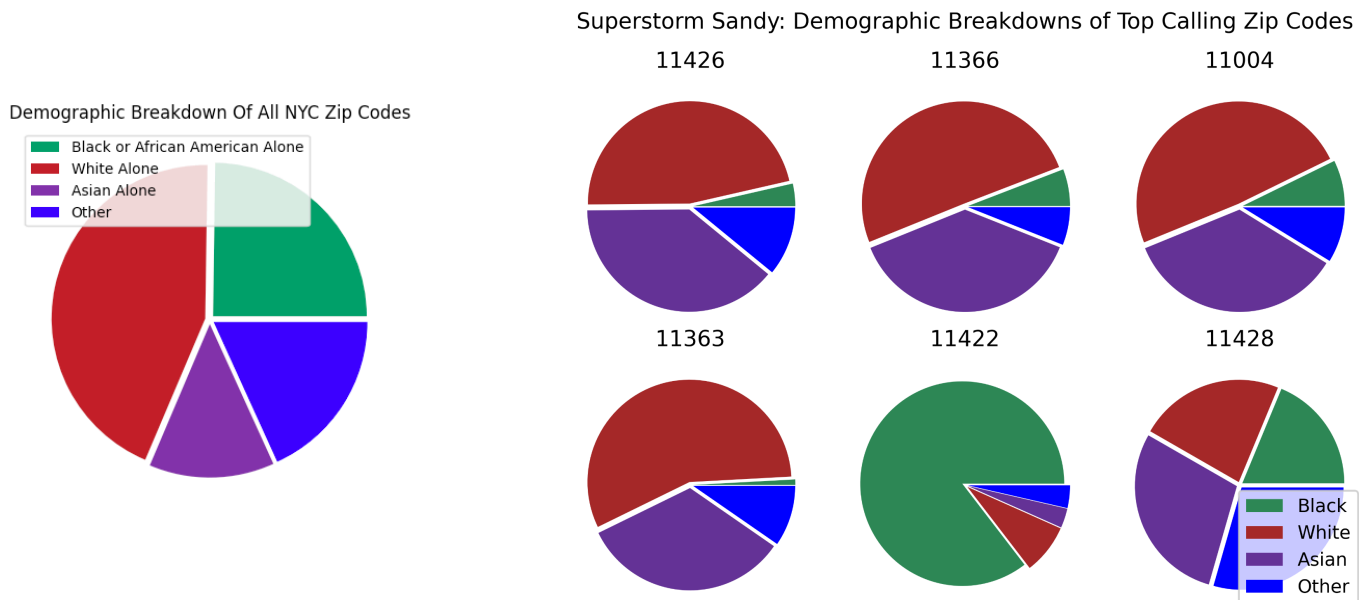
New York City's 311 service response program responds to requests for governmental and non-emergency services. All service requests are logged and available to the public on the NYC OpenData website. For every call, the database provides the date and time, complaint type, borough, zip code, address, and even geographic coordinates. These 311 calls can provide extremely important insight on who is experiencing flooding in the city and where flooding happens frequently. When working with these data, it is important to remember that access to 311 service can vary between communities. Whether a New York City resident doesn't own a cell phone, experiences a language barrier, or hasn't heard of 311 at all, there's many reasons this resource could be unusable. If certain communities are not receiving timely responses to their service requests, naturally, they may stop calling. Also, in the case of intense or catastrophic weather events, citizens who are actively fleeing their homes would not be able to call either. These limitations are important to keep in mind when analyzing the population demographics of flood service request callers.

SUPERSTORM SANDY:

On October 29th, 2012, an extratropical storm that developed from Hurricane Sandy as it traveled up the east coast made landfall in Atlantic City, NJ, pushing a catastrophic storm surge into the harbor of New York City. Over the course of 48 hours, the superstorm damaged over 69,000 residential units, destroyed 300 homes, and took the lives of 44 City residents. The intensity of Sandy was a wake-up call to many in the New York area. Major hurricanes are the costliest natural weather disasters, and the estimated cost of damage in NYC was \$19 billion. Nationally, FEMA estimated Sandy made a \$50 billion dent in property damage. Though the harm left in Sandy's wake was devastating, the event itself makes for a good case study, as it allows us to view how people respond to a large flooding event. Although the specific meteorological mechanics that [link climate change to increased risk from hurricanes](#) is not yet completely understood, it's clear that precipitation is increasing. The Clausius-Clapeyron equation, used for estimating the relationship between vapor pressure and temperature, dictates that for every one degree Celsius increase, the Earth's atmosphere can hold [7% more moisture](#). For tropical cyclones, a larger amount of moisture means not only greater amounts of precipitation, but in some cases also [greater intensity and rapid growth](#).



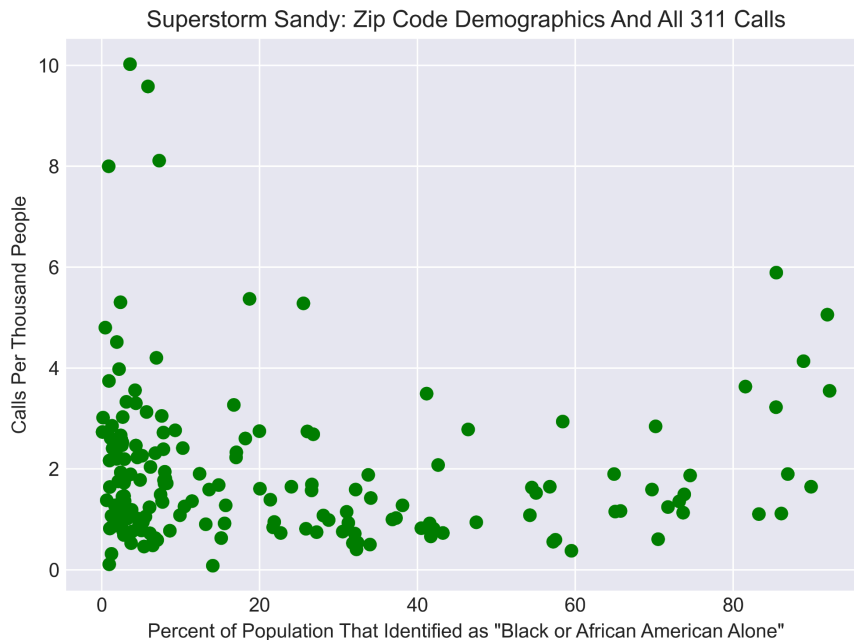
The most common complaint during the three total days of Sandy was by far calls about damaged or downed trees. As shown in the graph to the left, there were a lot of other complaints called in as well. In the top 15 categories, several pertain to flooding.



By using Census demographic data ([American Community Survey 5-year average 2010-2014](#)), we are able to visualize a basic breakdown

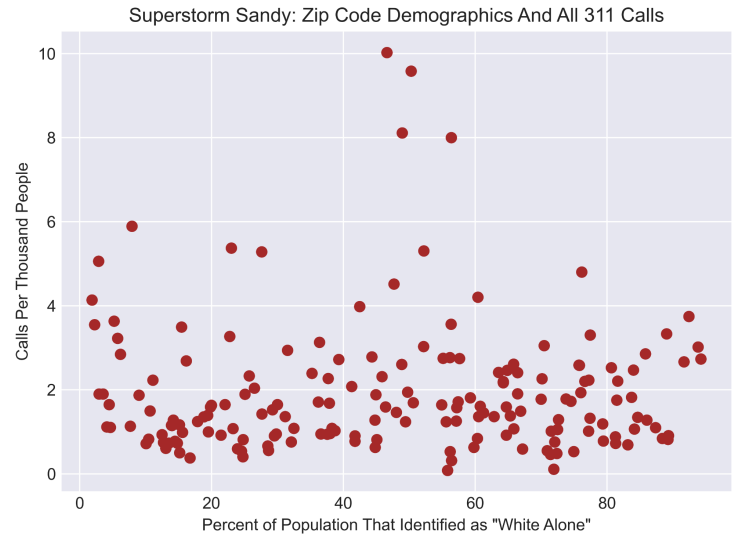
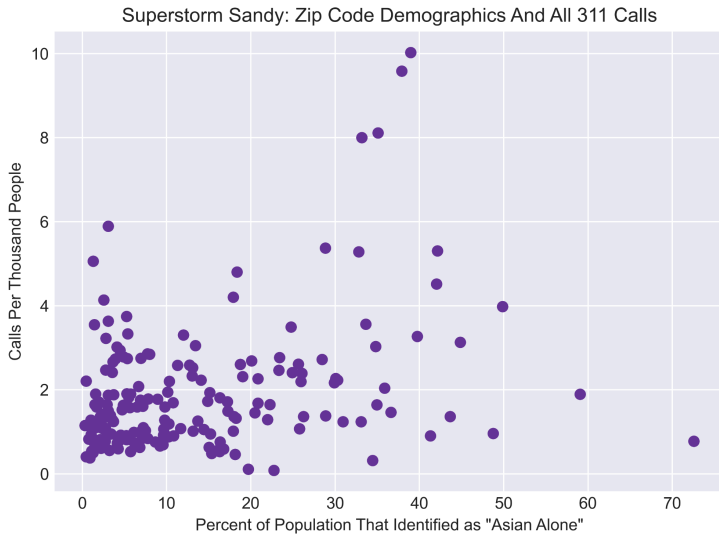
of the top-calling zip codes during Superstorm Sandy. It is important to note that race and ethnicity are both multidimensional and contextual, and are therefore challenging to minimize into distinct, representative categories. Census race categories reflect the political and racial thinking of the era in which they are created, and their origin as a part of the national census is rooted in racism. [A paper published by HV Strmic-Pawl, BH Jackson, and S. Garner \(2017\)](#) states that for the majority of the U.S. census' existence, "Whiteness and power motivated the addition and removal of racial groups, the names for racial groups, and the use of census officials to observe and record the race of participants."

Holding these truths in mind, the visualizations shown are made using census race categories in a utilitarian approach for a basic understanding of demographic makeup within different zip codes. Further work would need to be done in order to understand the nuances of these populations. The 'Other' category represents any person that did not mark 'Black or African American Alone', 'White Alone', or 'Asian Alone' on their census form, and includes people that may identify with more than one race. Comparing the zip codes with the most calls during Sandy and a breakdown of all zip codes within the city, there are stark differences, especially in the percent of the population who identified as Asian.



Distributing zip codes by the number of calls during Sandy and the percentage of the population that identified within a specific race category reveals interesting trends. The chart at left shows that there were a lot of calls during Superstorm Sandy coming from communities with >70% Black populations. We

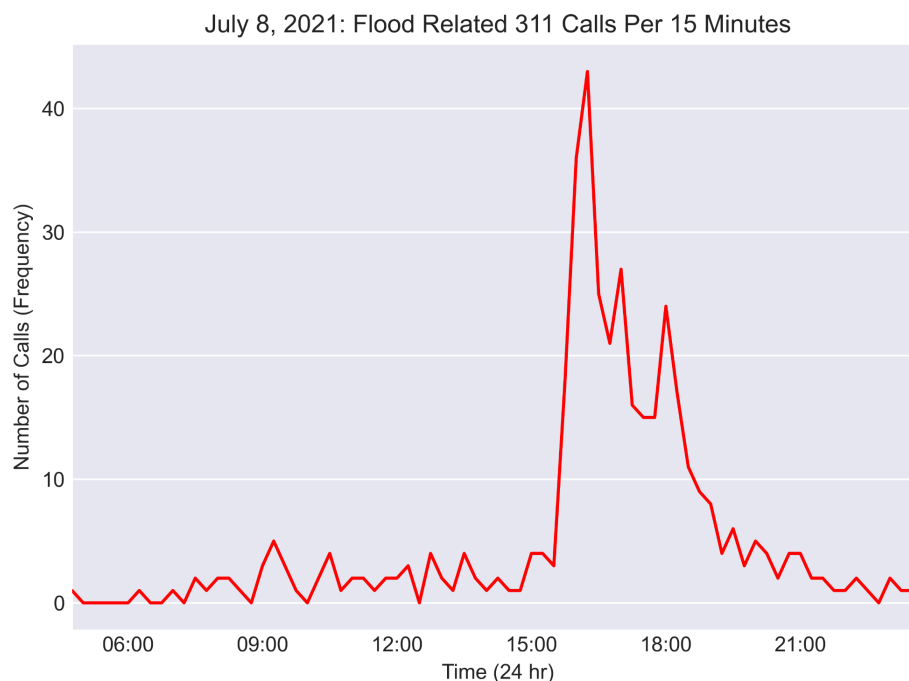
can compare this with the same distributions for Asian and White populations.



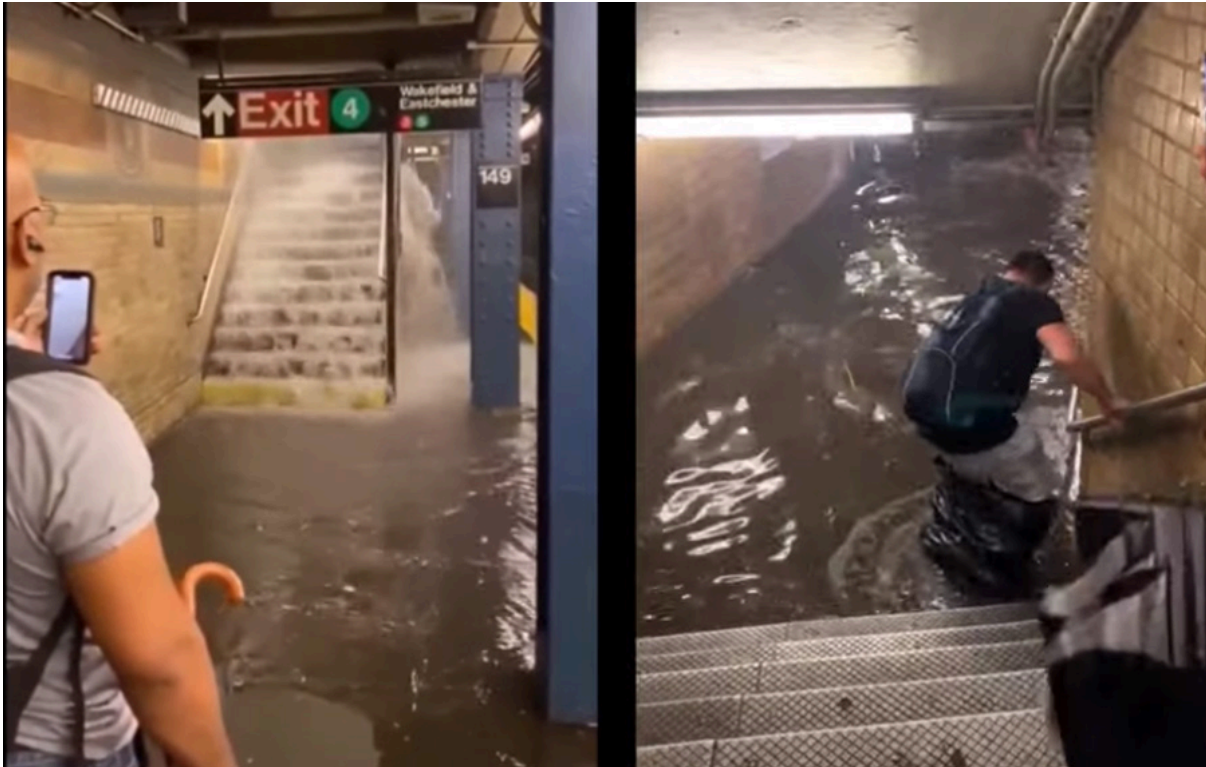
In the distribution of zip codes by Asian population there are fewer zip codes that have a >50% Asian population, and a few 30-40% Asian population zip codes that called very frequently during Sandy. Further research would need to be done in order to explain why some of these zip codes had such a low call rate.

2021 CLOUDBURST:

On July 8th, 2021, a cloudburst preceding Tropical Storm Elsa hit NYC creating deluges and flash floods in some streets and subway stations. Four inches of rain hit some areas of the city, most of which fell in the span of one hour between 3pm and 4pm.



When filtering the NYC OpenData 311 service dataset for flood-related calls, the following complaint types are included: 'Street Flooding', 'Sewer Backup', 'Highway Flooding', 'Manhole Overflow', 'Possible Water Main Break', 'Catch Basin Clogged/Flooding', and 'Excessive Water In Basement.'

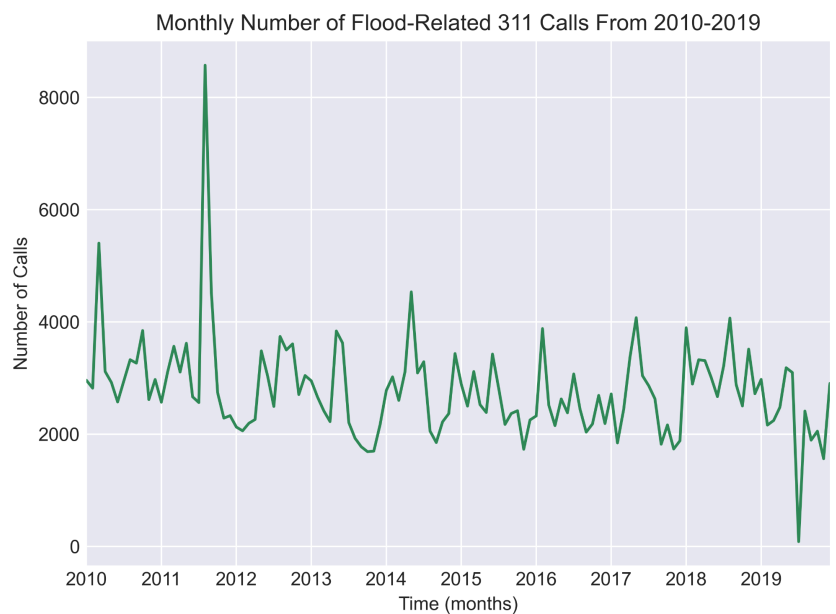


City residents' personal footage of subway station flooding during Tropical Storm Elsa

2010 TO 2019:

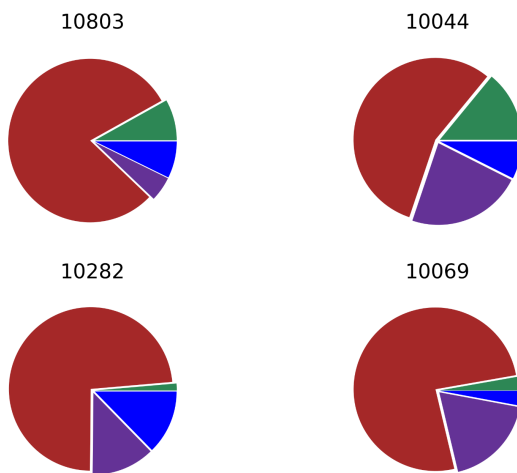
Expanding beyond these case study events to a larger time frame is the next step for this work.

A simple time series of flood related calls over the past decade shows that large amounts of flood-related 311 requests are not limited to extreme weather events, such as Superstorm Sandy or Tropical Storm Elsa.

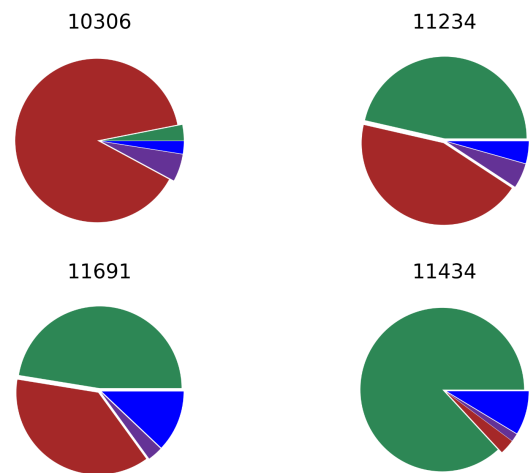


An interesting discovery in the demographic work is that normalizing the number of flood-related calls by zip code population does not accurately represent communities that were the most hard-hit by flooding. Normalizing by population emphasizes less populated zip codes, which may also have better access and understanding of 311 services. The demographic make ups of the top-calling zip codes change extremely when considering the total number of calls from a zip code, without normalizing by population.

Top Flood-Related 311 Calling Zip Code Demographics
From 2010 to 2019, Normalized by Population



Top Flood-Related 311 Calling Zip Code Demographics
From 2010 to 2019



FUTURE WORK:

More research needs to be done to fully understand the connection between flood risk and demographics in New York City. Part of this research should be the development of flood threshold levels for different communities, and determining how much precipitation an area can manage before flooding becomes threatening to businesses or lives. Determining flooding thresholds will give communities more information about their flood vulnerability and future flood risks. Finding new information on neighborhoods who are at-risk but are not included in the [FEMA National Flood Insurance Program hazard mapping](#) could mean the difference between a home saved and a home destroyed.

All visualizations were made using Python 3.9.1